

Muhammad Fayaz

PH.D. SCHOLAR IN AI AND COMPUTER VISION (EXPECTED GRADUATION OCTOBER) ·

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Research Interests

AI & Computer Vision

Advanced Machine Learning, Deep Learning, Anomaly Detection in Surveillance Video, Video Analysis, Medical Image Analysis, Activity Recognition, Image Processing, Object Detection, and Semantic Segmentation

Education

PhD Degree | CGPA : 4.38/4.5 (97.5%)

SEJONG UNIVERSITY

Department of Computer Science and Engineering

Thesis Title and Related Paper: TBD

Research Field: Computer Vision and Pattern Recognition (CVPR) Lab, Sejong University, with research interests in Remote Sensing, Anomaly Detection in Surveillance Video, Land Cover Mapping, Earth Observation, and Medical Images Analysis.

Sep. 2023 - Feb. 2027

Seoul, Republic of Korea

Master of Science in Computer Science | CGPA : 3.28/4.0 (80%)

CYPRUS INTERNATIONAL UNIVERSITY (CIU)

Department of Computer Science

Thesis Title and Related Paper: Brain Tumor Detection

Aug. 2019 - Sept. 2022

Lefkoshia, Turkish Republic of North

Cyprus (TRNC), Turkey

Bachelor of Science in Computer Science | CGPA : 3.12/4.0 (78%)

ISLAMIA COLLEGE PESHAWAR (ICP)

Department of Computer Science

Thesis Title and Related Paper: Online Donation System

Aug. 2014 - Sept. 2018

Peshawar, Pakistan

Work Experience

PhD. Researcher

COMPUTER VISION AND PATTERN RECOGNITION (CVPR) LAB, DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING, SEJONG UNIVERSITY

In this position, I conduct research, analyze data, write papers, manage projects, assist students, collaborate with researchers, and occasionally teach, preparing for a Ph.D. or an industry career.

Sep. 2023 - Feb 2027

Seoul, Republic of Korea

MS Student

MASTER OF SCIENCE IN COMPUTER SCIENCE, CYPRUS INTERNATIONAL UNIVERSITY (CIU)

Research and development of various industrial and research projects.

Aug. 2019 - Sep 2022

Lefkoshia, Turkish Republic of North

Cyprus (TRNC), Turkey

Academic Activity

Peer Review Records

- Expert Systems with Applications (Elsevier)
- Neurocomputing (Elsevier)
- Engineering Applications of Artificial Intelligence (Elsevier)
- International Journal of Applied Earth Observation and Geoinformation (Elsevier)
- IEEE Transactions on Neural Networks and Learning Systems (IEEE)

- Artificial Intelligence Review (Springer)
- Pattern Analysis and Applications (Springer)
- Scientific Reports(Springer)
- Pattern Analysis and Applications (Springer)
- Signal, Image and Video Processing (Springer)

Technical Skills

Research writing and visualization	[Expert]	Microsoft Word
	[Expert]	Microsoft Power Point
	[Intermediate]	Origin-pro
Deep learning frameworks	[Used]	TensorFlow (Neural networks, etc)
	[Used]	Keras
	[Pytorch]	Will explore in future
Programming and development skills	[Expert]	Python (OpenCV, Numpy, Scikit-learn, Pandas,Matplotlib)
	[Worked]	MATLAB (Image Processing), Anaconda Python, Jupyter notebook, Spyder, Latex, Code Ocean Capsule, and Kaggle.
	[Studied]	C++, JAVA

Participated Projects

#	Position	Project Title/Details	Funding Agency
1	Researcher	Development of Digital Quarantine and Operation Technologies for Creation of Safe Viewing Environment in Cultural Facilities	Korea Creative Content Service (Kocca, Korea Creative Content Agency)
2	Researcher	Developing System for Operating Unmanned Aerial Vehicle which aims at Automatic Monitoring Forest Disease	Korea Forest Service (KFS, Korea Forestry Promotion Agency)
3	Researcher	Development of digital quarantine and operation technology to create a safe viewing environment for cultural facilities	Electronics and Telecommunications Research Institute (ETRI)

Peer-Reviewed Journal Publications

GeoSwin: Hierarchical Representation Learning and a Large-Scale Earth Observation Benchmark

CVPR

MUHAMMAD FAYAZ, ZULFIQAR AHMAD KHAN, LIEN MINH DANG, AND HYEONJOON MOON*

Published: April 2026

[IF : 8.3, Rank Q1]

HQDL: A Hybrid Quantum-Deep Learning Framework for Real-Time Fire Detection Using UAV Remote Sensing

IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing

SUFYAN DANISH, WASEEM ULLAH, MUHAMMAD FAYAZ, L. MINH DANG, HYOUNG-KYU SONG, AND HYEONJOON MOON*

Accepted

[IF : 5.3, Rank Q1]

MedViTNet: A Hybrid Deep Learning Framework for Early and Accurate Detection of Retinal Diseases

Electronics

SUFYAN DANISH, MUHAMMAD FAYAZ, MUHAMMAD AYYAZ AKBAR KHAN, ASMAT NOREEN, AMIRMOHAMMAD MONTAZERI, CHUN-MING LAI, AND HYEONJOON MOON*

Revision 1 Submitted

[IF : 2.6, Rank Q2]

Hierarchical Optimization of Task-Adaptive Networks for Integrated Registration and Few-Shot Medical Image Segmentation

Biomedical Signal Processing and Control

MUHAMMAD FAYAZ, ZEESHAN ALI HAIDER, L. MINH DANG AND HYEONJOON MOON*

Published: April 2026

[IF : 4.9, Rank Q1]

Hierarchical Optimization of Task-Adaptive Networks for Integrated Registration and Few-Shot Medical Image Segmentation

ZEESHAN ALI HAIDER, MUHAMMAD FAYAZ, INAM ULLAH, MUNEEBA ALTAYEB, FAHAD ALBLEHAI AND AMR TOLBA*

[IF : 10.9, Rank Q1]

CALCNet: A Novel Cross-Module Attention Network for Efficient Land Cover Classification

MUHAMMAD FAYAZ, HIKMAT YAR*, WEIWEI JIANG, ANWAR HASSAN IBRAHIM, MUHAMMAD ISLAM, L. MINH DANG*

[IF : 4.1, Rank Q1]

A Comprehensive Approach for Image Quality Assessment Using Quality-Centric Embedding and Ranking Networks

ZEESHAN ALI HAIDER, SAREER UL AMIN, MUHAMMAD FAYAZ, FIDA MUHAMMAD KHAN, HYEONJOON MOON, AND SANGHYUN SEOE,*

[IF : 7.5, Rank Q1]

Efficient transformer-based semantic segmentation of colonic polyps using SegFormer

GUL E ARZU, MUHAMMAD FAYAZ, USMAN ALI, L MINH DANG, AND HYEONJOON MOON*

[IF : 6.5, Rank Q1]

PolySAGN: Hierarchical multi-scale representation learning with scale-specific attention for accurate polyp segmentation

WENQI ZHANG, YUE ZHANG, MUHAMMAD FAYAZ, L MINH DANG, TAN N NGUYEN AND HYEONJOON MOON*

[IF : 6.5, Rank Q1]

Enhancing pine wilt disease detection with synthetic data and external attention-based transformers

SAREER UL AMIN, YONGHOON JUNG, MUHAMMAD FAYAZ, BUMSOO KIM AND SANGHYUN SEO*

[IF : 8.0, Rank Q1]

Enhancing land cover classification via deep ensemble network

MUHAMMAD FAYAZ, L MINH DANG AND HYEONJOON MOON*

[IF : 7.2, Rank Q1]

DLAN: A dual attention network for effective land cover classification in remote sensing

MUHAMMAD FAYAZ, L MINH DANG AND HYEONJOON MOON*

[IF : 7.2, Rank Q1]

End-to-end plant disease detection using transformers with collaborative hybrid assignment trainings

YANFEN LI, MUHAMMAD FAYAZ, SUFYAN DANISH, HANXIANG WANG, TAN N NGUYEN AND L MINH DANG*

[IF : 6.6, Rank Q1]

Transformer-based detection of abnormal rice growth using drone-based multispectral imaging

YANFEN LI, L MINH DANG, HANXIANG WANG, MUHAMMAD FAYAZ, SUFYAN DANISH, JUNLIANG SHANG, HYOUNG-KYU SONG AND HYEONJOON MOON*

[IF : 8.9, Rank Q1]

Large language models in medical image analysis: a systematic survey and future directions

MUHAMMAD FAYAZ, BUSHRA UROOJ, L MINH DANG* AND KYUNG WON KIM*

[IF : 3.7, Rank Q1]

Land-cover classification using deep learning with high-resolution remote-sensing imagery

L MINH DANG, SUFYAN DANISH, ASMA KHAN, NUR ALAM, MUHAMMAD FAYAZ, DINH KHUONG NGUYEN, HYOUNG-KYU SONG AND HYEONJOON MOON*

[IF : 2.5, Rank Q2]

An efficient zero-labeling segmentation approach for pest monitoring on smartphone-based images

MUHAMMAD FAYAZ, JUNYOUNG NAM, L MINH DANG, HYOUNG-KYU SONG AND HYEONJOON MOON*

[IF : 5.5, Rank Q1]

Universal Image Segmentation with Arbitrary Granularity for Efficient Pest Monitoring

L. MINH DANG, SUFYAN DANISH, MUHAMMAD FAYAZ, ASMA KHAN, GUL E ARZU, HYOUNG-KYU SONG AND HYEONJOON MOON*

[IF : 5.5, Rank Q1]

IEEE Transaction on Consumer Electronics

Published: April 2026

Remote Sensing

Published: April 2026

Pattern Recognition

Published: December 2025

Neurocomputing

Published: December 2025

Neurocomputing

Published: December 2025

Engineering Applications of Artificial Intelligence

Published: November 2025

Knowledge-Based Systems

Published: December 2024

Knowledge-Based Systems

Published: June 2025

Applied Soft Computing

Published: October 2025

Computers and Electronics in Agriculture

Published: December 2025

Bioengineering

Published: July 2025

Applied Sciences

Published: February 2024

European Journal of Agronomy

Published: October 2024

Horticulturae

Published: December 2025

TumorNet: A hybrid lightweight framework for brain tumor classification and reasoning

HANXIANG WANG, MUHAMMAD ZAQEEM, **MUHAMMAD FAYAZ**, DEFU QIU, SAJJAD AHADZADEH, TAN N. NGUYEN* AND L. MINH DANG*

Information Sciences

Published: March 2026

[IF : 6.8, Rank Q1]

Masked Autoencoder-Based Vision Framework for Robust Fire Detection in Complex Environments

HANXIANG WANG, **MUHAMMAD FAYAZ**, AWAIS AHMAD, YANFEN LI, TAN N. NGUYEN* AND L. MINH DANG*

Process Safety and Environmental Protection

Published: October 2025

[IF : 7.8, Rank Q1]

Demystifying Artificial intelligence: A Systematic Review of Explainable Artificial Intelligence in Medical Imaging

MUHAMMAD FAYAZ, KIM HAGSONG, SUFYAN DANISH, L. MINH DANG, ABOLGHADEM SADEGHI NIARAKI, HYEONJOON MOON *

sensors

Published: March 2026

[IF : 3.5, Rank Q2]

SwinHCAD: A Robust Multi-Modality Segmentation Model for Brain Tumors Using Transformer and Channel-Wise Attention

SEYONG JIN, **MUHAMMAD FAYAZ**, L MINH DANG, HYOUNG-KYU SONG, HYEONJOON MOON*

Computers, Materials, & Continua

Published: 2026

[IF : 2.5, Rank Q3]

An Efficient and Robust Hand Gesture Recognition System of Sign Language Employing Finetuned Inception-V3 and Efficientnet-B0 Network

ADNAN HUSSAIN, SAREER UL AMIN, **MUHAMMAD FAYAZ**, AND SANGHYUN SEO*

Computer Systems Science and Engineering

Published: April. 2023

[IF : 4.397, Rank Q1]

Peer-Reviewed Journal Articles in Process

Optimized Vision Transformer with Shifted Patch and Local Attention for Robust Land Cover Mapping

MUHAMMAD FAYAZ, L MINH DANG AND HYEONJOON MOON*

Neurocomputing

Under Review

[IF : 6.5, Rank Q1]

Advanced Medical Representation Learning:

MUHAMMAD FAYAZ, L MINH DANG AND HYEONJOON MOON*

Neurocomputing

Under Review

[IF : 6.5, Rank Q1]

Generalizing Anomaly Detection in Medical Imaging through Visual-Language Model Adaptation

MUHAMMAD FAYAZ, SAREER UL AMIN, SUFYAN DANISH, SHER TAJ, L. MINH DANG, AND HYEONJOON MOON*

International Journal of Computer Vision

Under Review

[IF : 9.3, Rank Q1]

GeoSpikeFormer

MUHAMMAD FAYAZ, AMIR MONTAZERI, ZULFIQAR AHMAD KHAN, L. MINH DANG, AND HYEONJOON MOON*

IEEE Transaction on Image Processing

Under Review

[IF : 13.7, Rank Q1]

DeepGeo-Cover: A Comprehensive Review of Deep Learning Techniques and Applications in Land Cover Mapping and Remote Sensing

MUHAMMAD FAYAZ, SUFYAN DANISH, AMIR MONTAZERI, MINJONG CHEON, TURKI M. ALANAZI, L. MINH DANG, AND HYEONJOON MOON*

Information Fusion

Under Review

[IF : 15.5, Rank Q1]

FocalFireNet: A Lightweight FocalNet-Based Framework for Real-Time Fire Detection in UAV Remote Sensing Imagery

SUFYAN DANISH, **MUHAMMAD FAYAZ**, GULEARZU, L. MINH DANG, HYOUNG-KYUSONG AND HYEONJOON MOON*

Geo-spatial Information Science

Under Review

[IF :5.5, Rank Q1]

Conferences Attended

GeoSwin: Hierarchical Representation Learning and a Large-Scale Earth Observation

2026 **Benchmark**, Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Colorado, USA

TSF-CVGL: A Transformer-Based Semantic Framework for Cross-View Geolocalization,

2025 Proceedings of the Korea Information Science Society

Yeosu, Republic of Korea

Blockchain Integration for Secure and Transparent Satellite Remote Sensing Data Systems,

2025 IAAA

Da Nang, Vietnam

Honors & Awards

- 2024-27 **Research Grant**, Research funding support by Korea Government Ministry of Science and ICT (MSIT) under the National Research Foundation of Korea (NRF) grant. *Seoul, S. Korea*
- 2023-25 **Research Grant**, Research funding support by the Ministry of Culture, Sports, and Tourism under the Korea Creative Content Agency's Culture Technology R&D Program. *Seoul, S. Korea*
- 2019 **Laptop Award**, Prime Minister's Youth Program. *Pakistan*

References

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Seoul, Republic of Korea

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Seoul, Republic of Korea

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Prof. Hikmat Yar

Seoul, Republic of Korea

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